

Singular Perturbations, Moments and Descriptor Systems: Linear

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1 Linear Systems - Thinking about explicit solutions and moments

Consider the following linear time-invariant, SISO system:

$$\begin{aligned}\dot{x} &= Ax + Bu, \\ y &= Cx,\end{aligned}\tag{1}$$

$$\begin{aligned}\dot{w} &= Sw, \\ u &= Lw,\end{aligned}\tag{2}$$

Given initial conditions $x(0) = x_0$ and $w(0) = w_0$, the solution to the system is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}BL e^{S\tau}w_0 d\tau.\tag{3}$$

The output of the system is given by

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}BL e^{S\tau}w_0 d\tau.\tag{4}$$

If $\sigma(A) \cup \sigma(S) = \emptyset$, then there exists a unique solution Π to the Sylvester equation

$$A\Pi + BL = \Pi S.\tag{5}$$

Using this, we can substitute an expression for BL into the integral term for $x(t)$:

$$\begin{aligned}x(t) &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}(\Pi S - A\Pi)e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}\Pi S e^{S\tau}w_0 d\tau - \int_0^t e^{A(t-\tau)}A\Pi e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + (\Pi e^{St} - e^{At}\Pi)w_0.\end{aligned}\tag{6}$$

Forward Moments are defined to be

$$M_g = C\Pi\tag{7}$$

2 Problem Setup

Consider the following linear singularly perturbed system:

$$\begin{aligned}\dot{x}_1(t) &= A_{11}x_1(t) + A_{12}x_2(t) + B_1u(t), \\ \varepsilon\dot{x}_2(t) &= A_{21}x_1(t) + A_{22}x_2(t) + B_2u(t),\end{aligned}\tag{8}$$

where $x_1(t) \in \mathbb{R}^{n_1}$, $x_2(t) \in \mathbb{R}^{n_2}$, $u(t) \in \mathbb{R}^m$, and $0 < \varepsilon \ll 1$.

You can write this as a descriptor system:

$$E_1\dot{x} = Ax + Bu\tag{9}$$

where

$$E_1 = \begin{bmatrix} I_{n_1} & 0 \\ 0 & \varepsilon I_{n_2} \end{bmatrix}, \quad A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \quad (10)$$

The input signal is also a linear singularly perturbed system:

$$\begin{aligned} \dot{w}_1 &= S_{11}w_1 + S_{12}w_2 \\ \varepsilon \dot{w}_2 &= S_{21}w_1 + S_{22}w_2 \end{aligned} \quad (11)$$

where $w_1(t) \in \mathbb{R}^{r_1}$, $w_2(t) \in \mathbb{R}^{r_2}$, and $0 < \varepsilon \ll 1$. The input to the system is given by $u(t) = L_1w_1(t) + L_2w_2(t)$.

The input signal can also be written as a descriptor system like so:

$$E_2\dot{w} = Sw \quad (12)$$

where

$$E_2 = \begin{bmatrix} I_{r_1} & 0 \\ 0 & \varepsilon I_{r_2} \end{bmatrix}, \quad S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}. \quad (13)$$

The output of the system is given by

$$y(t) = C_1x_1(t) + C_2x_2(t). \quad (14)$$

3 Weierstrass Canonical Form

For a descriptor system of the form $E\dot{x} = Ax + Bu$, the matrix pencil (A, E) is said to be regular if there exists a complex number s such that $\det(sE - A) \neq 0$. If the matrix pencil (A, E) is regular, then there exist nonsingular matrices P and Q such that

$$PEQ = \begin{bmatrix} I_{n_1} & 0 \\ 0 & N \end{bmatrix}, \quad PAQ = \begin{bmatrix} J & 0 \\ 0 & I_{n_2} \end{bmatrix}, \quad (15)$$

where J is in Jordan canonical form, and N is a nilpotent matrix.

Let's assume (A, E_1) and (S, E_2) are regular. So for the singular perturbation system, we have

$$PE_1Q = \begin{bmatrix} I_{n_1} & 0 \\ 0 & N \end{bmatrix}, \quad PAQ = \begin{bmatrix} J_A & 0 \\ 0 & I_{n_2} \end{bmatrix}, \quad (16)$$

and

$$WE_2T = \begin{bmatrix} I_{r_1} & 0 \\ 0 & M \end{bmatrix}, \quad WST = \begin{bmatrix} J_S & 0 \\ 0 & I_{r_2} \end{bmatrix}, \quad (17)$$

where J_A and J_S are in Jordan canonical form, and $N^\nu = 0$ and $M^\mu = 0$ are nilpotent matrices. ν is the index of $\lambda E_1 - A$, and μ is the index of $\lambda E_2 - S$

Question: Can we find P and Q for this? How does this relate to A_{22} being Hurwitz?

We can rewrite the system with a coordinate transformation $x = Q\tilde{x}$ and $w = T\tilde{w}$, with $\tilde{x} = \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}$ and $\tilde{w} = \begin{bmatrix} \tilde{w}_1 \\ \tilde{w}_2 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} I_{n_1} & 0 \\ 0 & N \end{bmatrix} \begin{bmatrix} \dot{\tilde{x}}_1(t) \\ \dot{\tilde{x}}_2(t) \end{bmatrix} &= \begin{bmatrix} J_A & 0 \\ 0 & I_{n_2} \end{bmatrix} \begin{bmatrix} \tilde{x}_1(t) \\ \tilde{x}_2(t) \end{bmatrix} + \tilde{B}(L_1\tilde{w}_1(t) + L_2\tilde{w}_2(t)), \\ \begin{bmatrix} I_{r_1} & 0 \\ 0 & M \end{bmatrix} \begin{bmatrix} \dot{\tilde{w}}_1(t) \\ \dot{\tilde{w}}_2(t) \end{bmatrix} &= \begin{bmatrix} J_S & 0 \\ 0 & I_{r_2} \end{bmatrix} \begin{bmatrix} \tilde{w}_1(t) \\ \tilde{w}_2(t) \end{bmatrix}, \\ y(t) &= \tilde{C}_1\tilde{x}_1(t) + \tilde{C}_2\tilde{x}_2(t), \end{aligned} \quad (18)$$

Question: When we set epsilon to 0, are the pencils no longer regular?

Moments obtained from these systems, then take epsilon to 0 and find convert back to original should give the same Pi. Might not be necessary to prove this bit. But if we say they are regular, then we can apply the initial value conditions for existence of solutions.

Consistent initial values $\implies$ Error is 0 for all t

Inconsistent initial values $\implies$ Neighbourhood around boundary layer?

4 Back to Finite Epsilon

For finite epsilon, E_2 is no longer singular. The overall Sylvester equation then becomes:

$$A\Pi + BL = E_1\Pi E_2^{-1}S \quad (19)$$

The main idea is that although $\lim_{\epsilon \rightarrow 0} E_2^{-1}$ is undefined, $\lim_{\epsilon \rightarrow 0} E_1\Pi E_2^{-1}$ is defined.

4.1 Explicit Solution for Finite Epsilon

For finite ϵ , we can write the explicit solution for (9) and (10) as such:

$$w(t) = \exp(E_2^{-1}St)w_0 \quad (20)$$

$$x(t) = \exp(E_1^{-1}At)x_0 + (\Pi \exp(E_2^{-1}St) - \Pi \exp(E_1^{-1}At)\Pi)w_0 \quad (21)$$

4.2 Boundary Layer Model

When we set epsilon to zero, the resulting boundary layer model will be:

$$\dot{w}_1 = (S_{11} - S_{12}S_{22}^{-1}S_{21})w_1 \quad (22)$$

$$\dot{x}_1 = (A_{11} - A_{12}A_{22}^{-1}A_{21})x_1 + (B_1 - A_{12}A_{22}^{-1}B_2)(L_1 - L_2S_{22}^{-1}S_{21})w_1 \quad (23)$$

Corresponding Π for (23) is $\Pi_{11}^{(0)}$ as we can show.

References